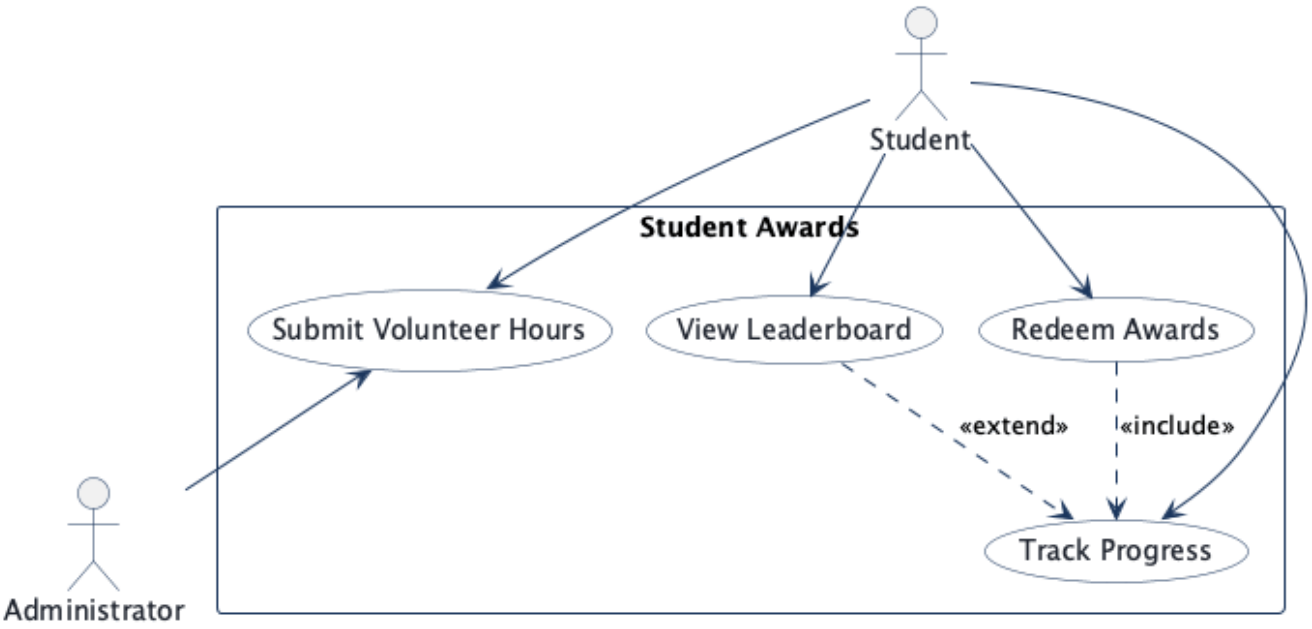
COMP 3613 Assignment 2

Individual report

Field	Value
Student name	Dana Ramtahal
Student ID	816047190
Deployed application	https://comp3613a1-pln2.onrender.com
GitHub repository	https://github.com/DanaRPR/comp3613a1

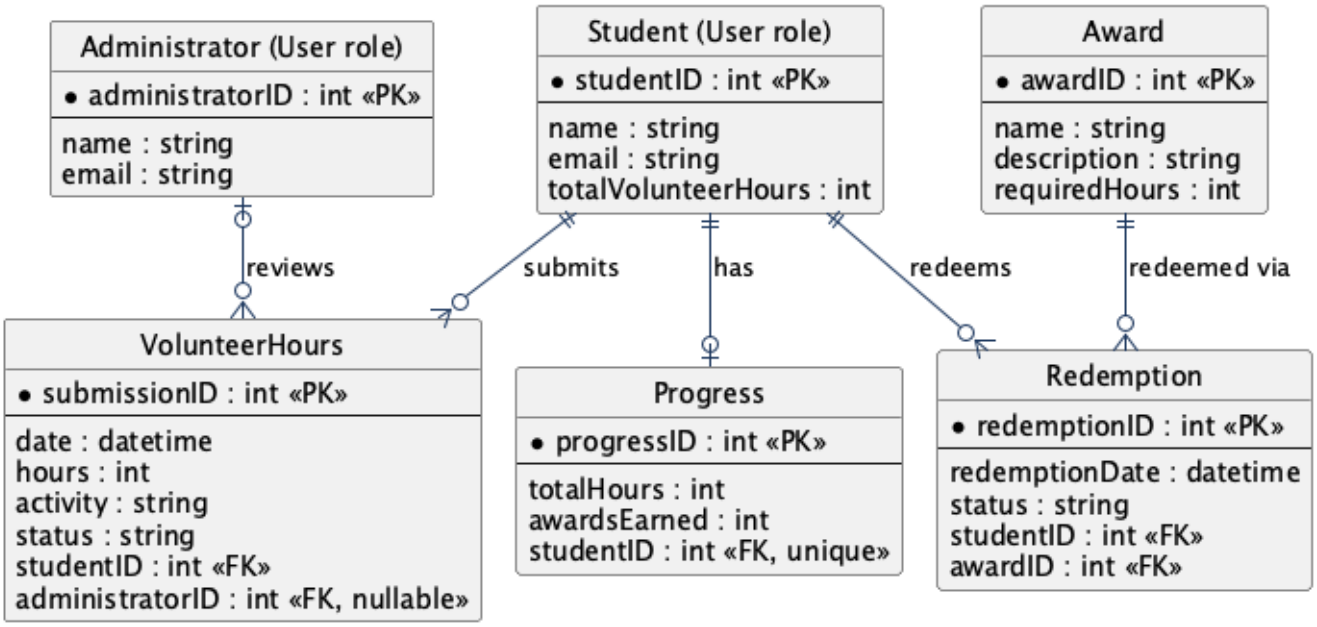
Project Description

Student Awards is a web-based incentive system for student volunteering. Students submit volunteer hours for administrator review, track approved hours and leaderboard rankings, and redeem awards when they meet the required-hour thresholds. Administrators approve or reject pending submissions; approved hours update student totals and progress. The application uses FastAPI, SQLModel, server-rendered HTML views, and a layered route/service/repository/model structure.



Use Case Diagram

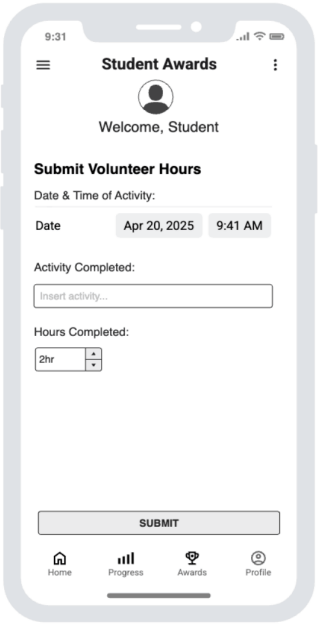
Students submit volunteer hours, track progress, view the leaderboard, and redeem awards. Administrators review submissions by approving or rejecting them. Viewing the leaderboard is optional after tracking progress; redeeming an award includes checking progress.

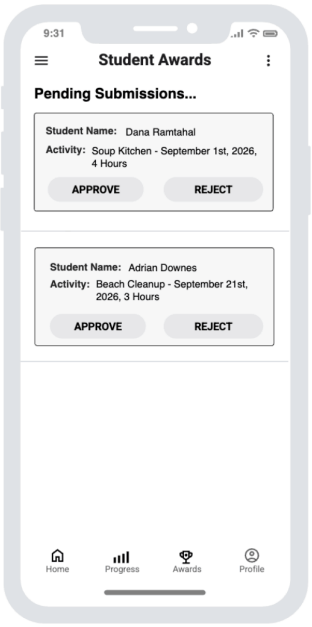


Entity Relationship Diagram

Student and Administrator denote role-specific User records in the implementation. A student may have no Progress row until their first approved submission; each existing Progress row belongs to one student. Pending VolunteerHours rows have no reviewing administrator; after approval or rejection, the row references the reviewing administrator.

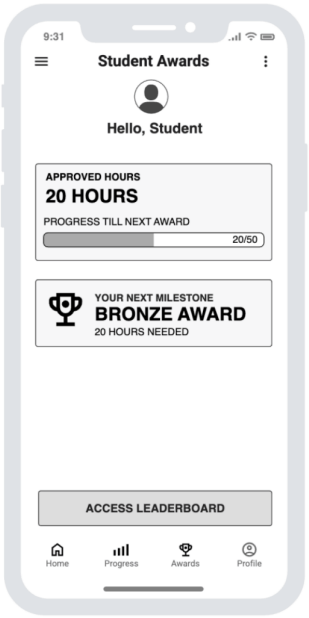
SUBMIT VOLUNTEER HOURS:

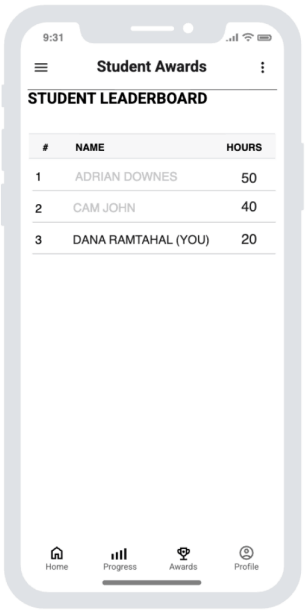




Application Wireframe - Submit Volunteer Hours

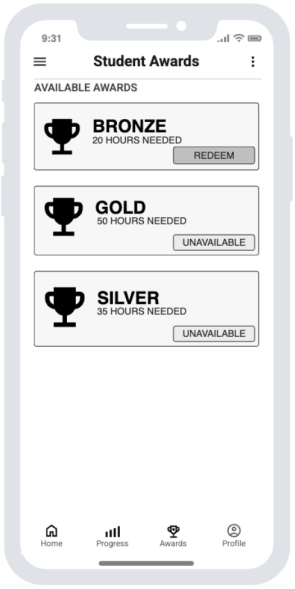
TRACK PROGRESS AND VIEW LEADERBAORD:

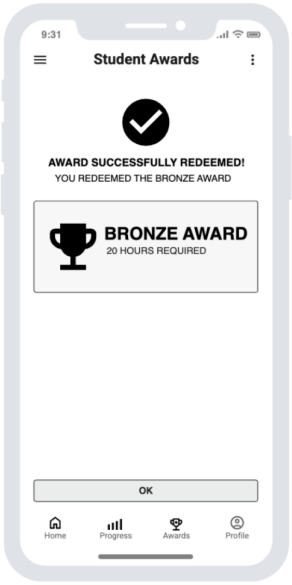




Application Wireframe - Track Progress and View Leaderboard

REDEEM AWARDS:





Application Wireframe - Redeem Awards

Test Plans

User Acceptance Test Plan and Completed Report

The scenarios are based on the application wireframes. Actual results and PASS outcomes below are the recorded UAT results.

UAT-01 - Submit Volunteer Hours

• **Role:** Student
• **Expected:** The student sees a success message and the administrator's pending list shows the new submission.
• **Actual:** The student successfully submitted 5 volunteer hours for a community clean-up activity and received a success message.
• **Result:** PASS

UAT-02 - Administrator Approves Volunteer Hours

• **Role:** Administrator, then Student
• **Expected:** The submission is no longer pending and approved hours increase by the submitted amount.
• **Actual:** The administrator approved the 5-hour Community clean-up submission. The submission disappeared from the pending submissions list.
• **Result:** PASS

UAT-03 - Reject a Pending Submission

• **Role:** Administrator, then Student
• **Expected:** The submission is no longer pending and the student's approved-hours total does not increase because of the rejection.
• **Actual:** The administrator rejected the pending volunteer-hours submission. It disappeared from the pending submissions list, leaving no pending submissions.
• **Result:** PASS

UAT-04 - View Progress and Next Milestone

• **Role:** Student
• **Expected:** The page displays the student's current approved hours and, when applicable, the next unredeemed award milestone.
• **Actual:** The student's total volunteer hours increased to 10 hours after the administrator approved the 5-hour Community clean-up submission.
• **Result:** PASS

UAT-05 - View Leaderboard

• **Role:** Student
• **Expected:** Students appear in descending approved-hours order; the administrator is excluded and the signed-in student is marked.
• **Actual:** The leaderboard displayed bob in first place with 10 volunteer hours. Other students with 0 hours appeared below him.
• **Result:** PASS

UAT-06 - Redeem an Eligible Award

• **Role:** Student
• **Expected:** The eligible award can be redeemed and is then displayed as redeemed.
• **Actual:** The student reached 20 approved volunteer hours, making the Bronze Award eligible for redemption. The student clicked Redeem and the application displayed 'Redeemed', confirming the award was successfully redeemed.
• **Result:** PASS

UAT-07 - Attempt to Redeem an Unavailable Award

• **Role:** Student
• **Expected:** An ineligible or already redeemed award cannot be redeemed.
• **Actual:** The Bronze, Silver and Gold awards were marked as unavailable because the student had only 10 approved volunteer hours. The redemption buttons were disabled, preventing the student from attempting to redeem an award.
• **Result:** PASS

Separate award test scenarios: UAT-06 and UAT-07 used different test states. For UAT-06, the student had 20 approved volunteer hours, making the Bronze Award eligible for redemption. For UAT-07, a separate test state had 10 approved volunteer hours, so the awards remained unavailable.

The UAT outcomes are the recorded user observations. The automated test screenshots in this report are separate from the UAT evidence.

Unit Test Plan

The application seeds Bronze, Silver, and Gold with thresholds of 20, 35, and 50 approved hours, respectively. Test descriptions that use other values refer to test-specific awards, not the seeded award catalog. The unit tests use fake repositories, session objects, templates, and services. They do not use the SQLite integration fixture or connect to the configured application database. The descriptions below reflect the assertions made by the current test functions.

• **test_submit_hours_adds_pending_submission()** - Checks pending status, student ID, hours, activity, and that the fake repository stores the returned submission; it does not assert the date.
• **test_approve_updates_submission_and_totals()** - Checks approved status, administrator ID, student total 10->18, progress total 15->23, and a fake-session add; it does not assert commit or persisted state.
• **test_approve_raises_value_error_for_missing_submission()** - Expects ValueError matching Submission not found.
• **test_reject_marks_submission_rejected()** - Checks rejected status, administrator ID, and a fake-session add; it does not assert unchanged student/progress totals.
• **test_get_available_awards_marks_eligible_and_not_redeemed()** - Checks an award above its threshold is available and not already redeemed.
• **test_get_available_awards_marks_redeemed_and_ineligible()** - Checks an under-threshold award that is also listed as redeemed is unavailable and already redeemed; the two conditions are combined.
• **test_redeem_raises_for_missing_award()** - Expects ValueError matching Award not found.
• **test_redeem_raises_for_duplicate_redemption()** - Expects ValueError matching already redeemed when the award ID is already redeemed.
• **test_get_leaderboard_assigns_rank_and_current_student()** - Checks first-row rank, name, hours, and current-student flag, plus the second row's non-current flag; fake rows are already ordered.
• **test_get_progress_returns_next_award_and_threshold()** - Checks the returned progress object, next unredeemed test award, and test-specific threshold of 25 hours (not the seeded Silver threshold).
• **test_submit_action_renders_value_error_message()** - A fake service raises ValueError("invalid submission"); checks HTTP 400, the submit_hours.html template, and the error context text.

Integration Test Plan

Integration tests use SQLModel records and the shared session fixture, which creates a temporary SQLite database under tmp_path; they do not connect to the deployed database. The descriptions below reflect the assertions made by the current test functions.

• **test_submit_hours_creates_pending_submission()** - Reads the created record and checks existence, pending status, student ID, and 5 hours; not date or activity.
• **test_approve_submission_updates_student_and_progress()** - Checks approved status, stored student total of 8, progress total of 8, and zero awards earned; not administrator ID.
• **test_approve_submission_creates_progress_when_missing()** - Checks approval and creation of a Progress row with 8 hours and zero awards earned.
• **test_approve_submission_rejects_duplicate_review()** - Checks a second approval raises ValueError matching already been reviewed; not post-error totals.
• **test_approve_missing_submission_raises_value_error()** - Expects ValueError matching Submission not found.
• **test_reject_submission_sets_rejected_status()** - Checks rejected status and administrator ID 7; not unchanged totals.
• **test_reject_missing_submission_raises_value_error()** - Expects ValueError matching Submission not found.
• **test_pending_submissions_include_student_names()** - Checks two rows, expected student names, and expected pending activities; it does not directly assert the reviewed row's absence by ID/status.
• **test_get_available_awards_eligible_and_unredeemed()** - Checks an above-threshold, unredeemed award is available and not already redeemed.
• **test_get_available_awards_marks_ineligible_and_redeemed()** - Checks an under-threshold award that is also redeemed is unavailable and already redeemed; these conditions are combined.
• **test_redeem_award_creates_redemption_and_updates_progress()** - Checks returned redemption status/student ID and persisted Progress.awardsEarned == 1; not an independent query for the redemption row.
• **test_redeem_award_does_not_create_missing_progress()** - Checks returned redemption is completed and no Progress row exists afterward; not an independent query for a saved Redemption row.
• **test_redeem_award_rejects_insufficient_hours()** - Expects ValueError matching You need; not database state afterward.
• **test_redeem_award_rejects_duplicate()** - Checks a second redemption raises ValueError matching already redeemed; not the number of redemption rows afterward.
• **test_leaderboard_excludes_admins_and_orders_students_by_hours()** - Checks returned names/order, first-row rank, and current-student flag; seeded names/order exclude the admin, but not all displayed hour values or tie handling.
• **test_get_progress_returns_next_unredeemed_award()** - Checks progress total 30 and that the next award and threshold are None when the only award is redeemed.
• **test_get_progress_returns_first_unredeemed_award()** - Checks the test award named Silver is selected after Bronze is redeemed and the test-specific threshold is 25 hours (not the seeded Silver threshold of 35); not the returned progress object or total.

Implementation

The application implements the Student Awards student and administrator workflows using FastAPI views and SQLModel-backed services and repositories. Authenticated student routes cover volunteer-hour submission, progress, leaderboard, and awards. Administrator-only routes cover pending-submission review. Service logic applies approval, progress, leaderboard, award-eligibility, and redemption rules; routers handle request binding, access dependencies, rendered responses, and redirects.

The workflow views use GET for displaying pages and POST for submissions and actions. Form actions that complete through navigation redirect back to the relevant page. The student and administrator pages are protected by their respective authentication dependencies.

Deployed Application

https://comp3613a1-pln2.onrender.com

Demo accounts: bob / bobpass (student) and admin / adminpass (administrator).

GitHub Repository

https://github.com/DanaRPR/comp3613a1

Test Implementation and API Test Collection

• **Unit tests:** 11 pytest tests in tests/unit/.
• **Integration tests:** 17 pytest tests in tests/integration/.
• **CLI command:** PYTHONPATH=. PYTHONDONTWRITEBYTECODE=1 .venv/bin/pytest -p no:cacheprovider -q
• **2026-11-10:** 11 unit + 17 integration tests; 28 passed, 0 failed, 0 skipped.
• **Postman collection:** student-awards-api-tests.postman_collection.json, exported as a Postman Collection v2.1 JSON file with 21 requests. It covers login and registration, student and administrator pages/actions, form validation, and access-control responses.
• **Postman run evidence:** The saved collection-run screenshots show 24 passed, 0 failed, and 0 errors. Postman was not rerun for this PDF revision.

```
(.venv) danaramtahal@Danas-Macbook comp3613a1 % PYTHONPATH=. pytest tests/unit -v
========================================= test session starts ==========================================
platform darwin -- Python 3.14.0, pytest-9.1.1, pluggy-1.6.0 -- /Users/danaramtahal/comp3613a1/.venv/bin/python
cachedir: .pytest_cache
rootdir: /Users/danaramtahal/comp3613a1
configfile: pyproject.toml
plugins: asyncio-1.4.0, anyio-4.15.1
asyncio: mode=Mode.AUTO, debug=False, asyncio_default_fixture_loop_scope=None, asyncio_default_test_loop_scope=function
collected 11 items

tests/unit/test_award_service_unit.py::test_get_available_awards_marks_eligible_and_not_redeemed PASSED [  9%]
tests/unit/test_award_service_unit.py::test_get_available_awards_marks_redeemed_and_ineligible PASSED [ 18%]
tests/unit/test_award_service_unit.py::test_redeem_raises_for_missing_award PASSED       [ 27%]
tests/unit/test_award_service_unit.py::test_redeem_raises_for_duplicate_redemption PASSED [ 36%]
tests/unit/test_leaderboard_service_unit.py::test_get_leaderboard_assigns_rank_and_current_student PASSED [ 45%]
tests/unit/test_leaderboard_service_unit.py::test_get_progress_returns_next_award_and_threshold PASSED [ 54%]
tests/unit/test_submit_hours_router_unit.py::test_submit_action_renders_value_error_message PASSED [ 63%]
tests/unit/test_volunteer_hours_service_unit.py::test_submit_hours_adds_pending_submission PASSED [ 72%]
tests/unit/test_volunteer_hours_service_unit.py::test_approve_updates_submission_and_totals PASSED [ 81%]
tests/unit/test_volunteer_hours_service_unit.py::test_approve_raises_value_error_for_missing_submission PASSED [ 90%]
tests/unit/test_volunteer_hours_service_unit.py::test_reject_marks_submission_rejected PASSED [100%]

========================================== 11 passed in 0.16s ==========================================
(.venv) danaramtahal@Danas-Macbook comp3613a1 %
```

Unit Test Report

```
(.venv) danaramtahal@Danas-Macbook comp3613a1 % PYTHONPATH=. pytest tests/integration -v
============================================================ test session starts =============================================================
platform darwin -- Python 3.14.0, pytest-9.1.1, pluggy-1.6.0 -- /Users/danaramtahal/comp3613a1/.venv/bin/python
cachedir: .pytest_cache
rootdir: /Users/danaramtahal/comp3613a1
configfile: pyproject.toml
plugins: asyncio-1.4.0, anyio-4.15.1
asyncio: mode=Mode.AUTO, debug=False, asyncio_default_fixture_loop_scope=None, asyncio_default_test_loop_scope=function
collected 17 items

tests/integration/test_award_service_integration.py::test_get_available_awards_eligible_and_unredeemed PASSED                  [  5%]
tests/integration/test_award_service_integration.py::test_get_available_awards_marks_ineligible_and_redeemed PASSED            [ 11%]
tests/integration/test_award_service_integration.py::test_redeem_award_creates_redemption_and_updates_progress PASSED          [ 17%]
tests/integration/test_award_service_integration.py::test_redeem_award_does_not_create_missing_progress PASSED                 [ 23%]
tests/integration/test_award_service_integration.py::test_redeem_award_rejects_insufficient_hours PASSED                       [ 29%]
tests/integration/test_award_service_integration.py::test_redeem_award_rejects_duplicate PASSED                                [ 35%]
tests/integration/test_leaderboard_service_integration.py::test_leaderboard_excludes_admins_and_orders_students_by_hours PASSED [ 41%]
tests/integration/test_leaderboard_service_integration.py::test_get_progress_returns_next_unredeemed_award PASSED             [ 47%]
tests/integration/test_leaderboard_service_integration.py::test_get_progress_returns_first_unredeemed_award PASSED            [ 52%]
tests/integration/test_volunteer_hours_service_integration.py::test_submit_hours_creates_pending_submission PASSED            [ 58%]
tests/integration/test_volunteer_hours_service_integration.py::test_approve_submission_updates_student_and_progress PASSED    [ 64%]
tests/integration/test_volunteer_hours_service_integration.py::test_approve_submission_creates_progress_when_missing PASSED   [ 70%]
tests/integration/test_volunteer_hours_service_integration.py::test_approve_submission_rejects_duplicate_review PASSED        [ 76%]
tests/integration/test_volunteer_hours_service_integration.py::test_approve_missing_submission_raises_value_error PASSED      [ 82%]
tests/integration/test_volunteer_hours_service_integration.py::test_reject_submission_sets_rejected_status PASSED             [ 88%]
tests/integration/test_volunteer_hours_service_integration.py::test_reject_missing_submission_raises_value_error PASSED      [ 94%]
tests/integration/test_volunteer_hours_service_integration.py::test_pending_submissions_include_student_names PASSED          [100%]

============================================================= 17 passed in 0.22s =============================================================
(.venv) danaramtahal@Danas-Macbook comp3613a1 %
```

Integration Test Report

GET Admin - View Review Page

http://127.0.0.1:8000/admin

200 • 3 ms • 6.872 KB 1

PASS admin cookie grants access to review page

POST Admin - Approve Submission (redirect behavior)

http://127.0.0.1:8000/admin/approve/1

200 • 2 ms • 6.872 KB 1

PASS approval attempt ends on admin review page after redirect

POST Admin - Reject Submission (redirect behavior)

http://127.0.0.1:8000/admin/reject/1

200 • 2 ms • 6.872 KB 1

PASS rejection attempt ends on admin review page after redirect

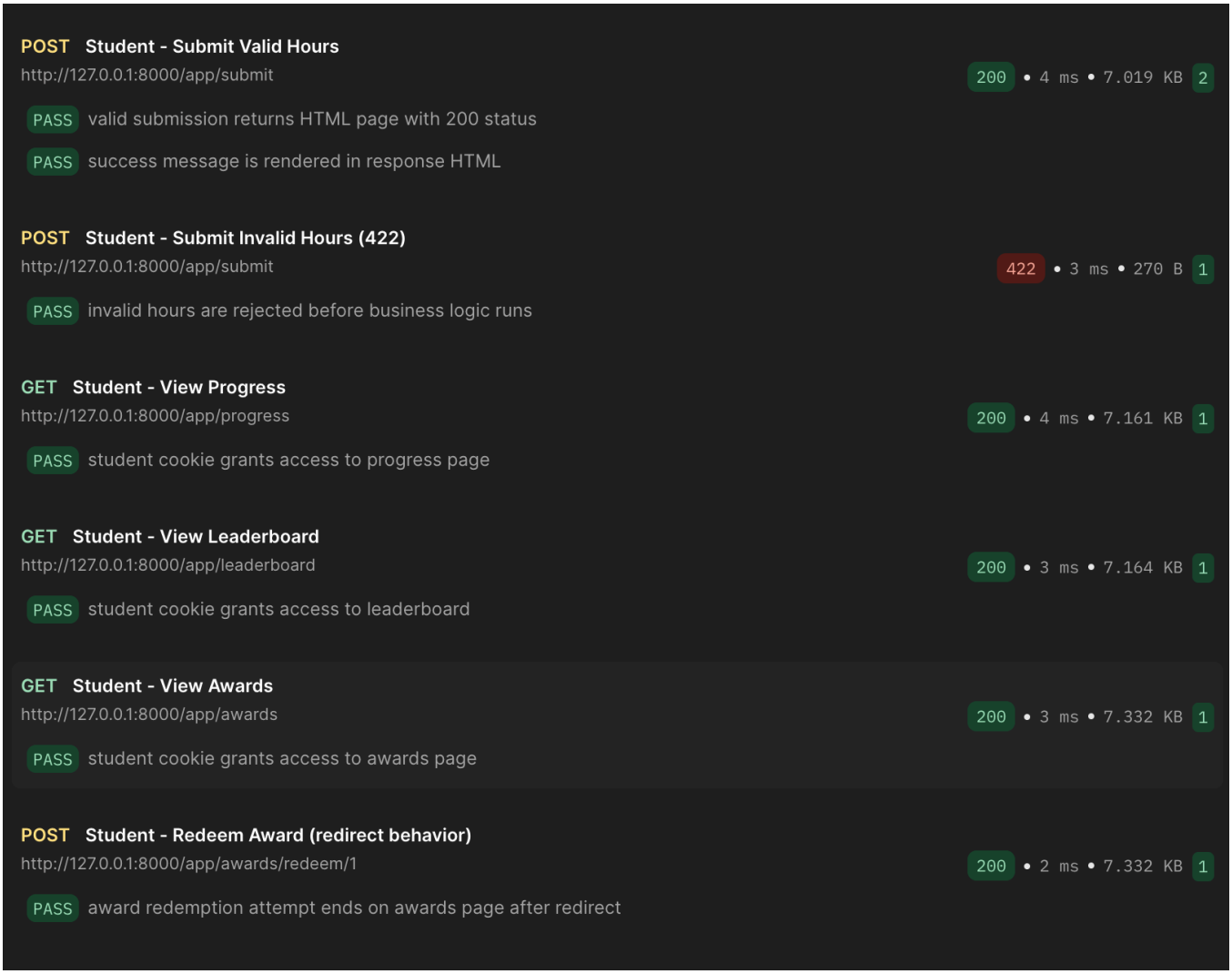
GET Access Control - Protected route without cookie

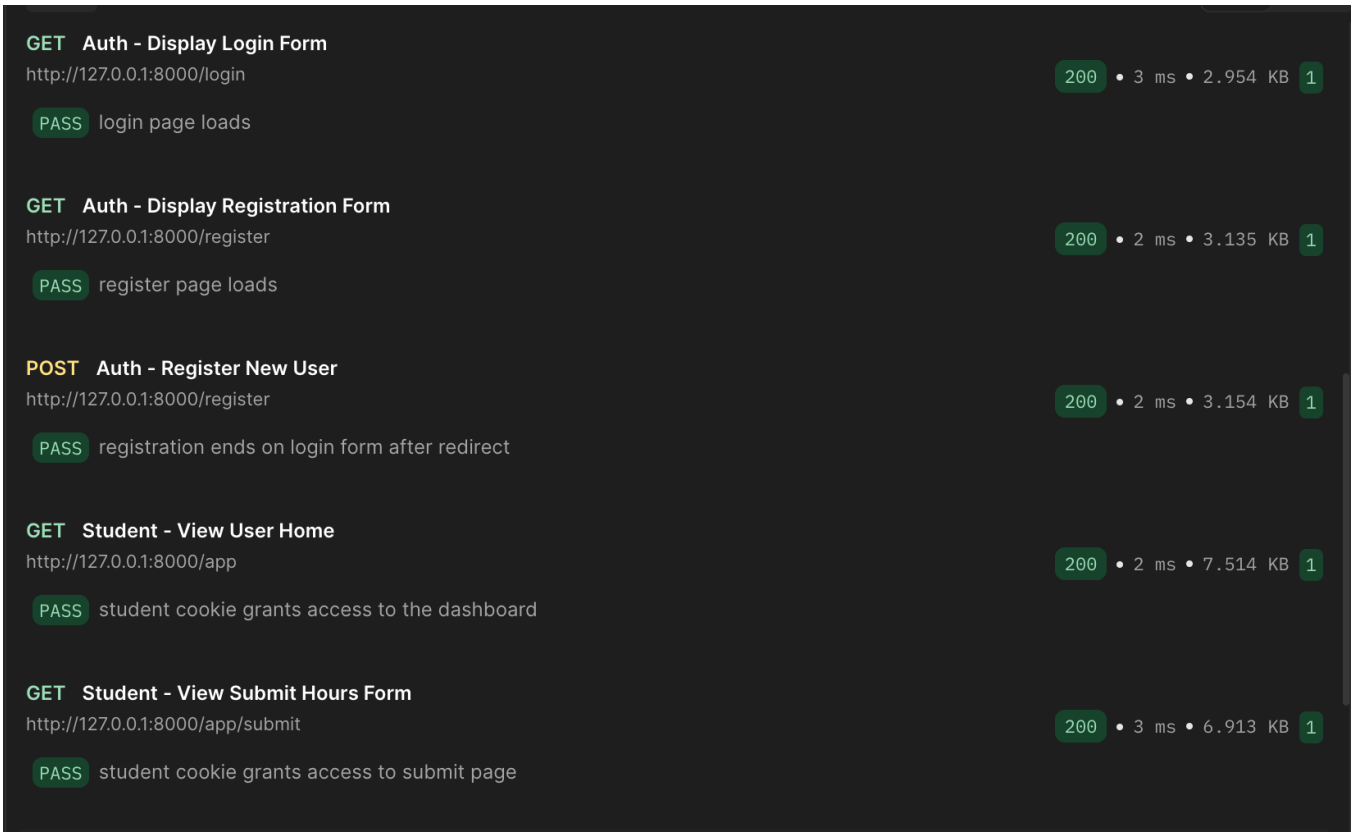
http://127.0.0.1:8000/app/progress

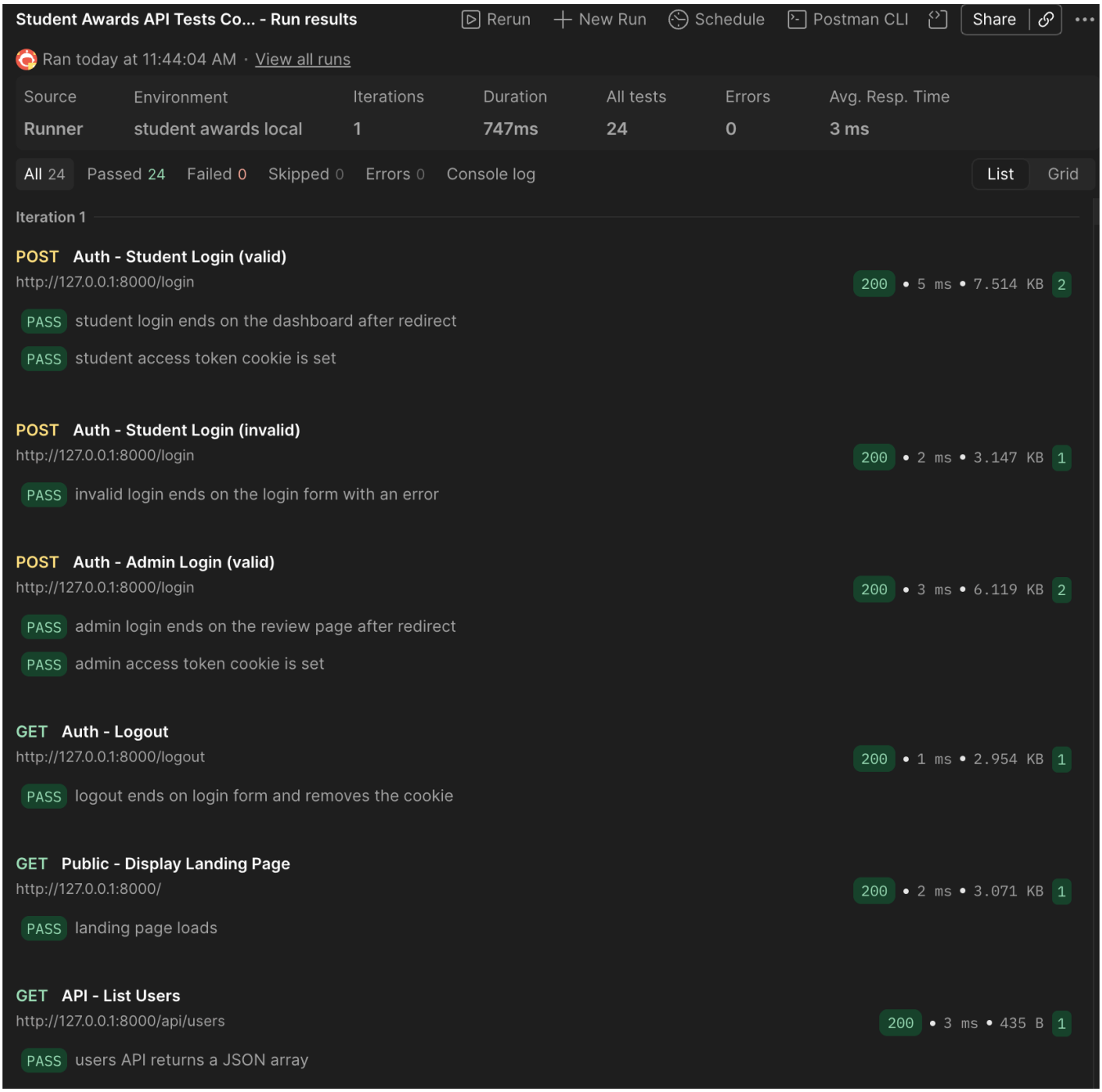
200 • 2 ms • 3.926 KB 1

PASS unauthenticated request renders the unauthorized page

Postman Collection Run Report




GET Auth - Display Login Form
http://127.0.0.1:8000/login
200 • 3 ms • 2.954 KB 1
PASS login page loads
GET Auth - Display Registration Form
http://127.0.0.1:8000/register
200 • 2 ms • 3.135 KB 1
PASS register page loads
POST Auth - Register New User
http://127.0.0.1:8000/register
200 • 2 ms • 3.154 KB 1
PASS registration ends on login form after redirect
GET Student - View User Home
http://127.0.0.1:8000/app
200 • 2 ms • 7.514 KB 1
PASS student cookie grants access to the dashboard
GET Student - View Submit Hours Form
http://127.0.0.1:8000/app/submit
200 • 3 ms • 6.913 KB 1
PASS student cookie grants access to submit page


Student Awards API Tests Co... - Run results
Rerun
New Run
Schedule
Postman CLI
Share
Ran today at 11:44:04 AM · View all runs
Source
Environment
Iterations
Duration
All tests
Errors
Avg. Resp. Time
Runner
student awards local
1
747ms
24
0
3 ms
All 24
Passed 24
Failed 0
Skipped 0
Errors 0
Console log
List
Grid
Iteration 1
POST Auth - Student Login (valid)
http://127.0.0.1:8000/login
200 • 5 ms • 7.514 KB 2
PASS student login ends on the dashboard after redirect
PASS student access token cookie is set
POST Auth - Student Login (invalid)
http://127.0.0.1:8000/login
200 • 2 ms • 3.147 KB 1
PASS invalid login ends on the login form with an error
POST Auth - Admin Login (valid)
http://127.0.0.1:8000/login
200 • 3 ms • 6.119 KB 2
PASS admin login ends on the review page after redirect
PASS admin access token cookie is set
GET Auth - Logout
http://127.0.0.1:8000/logout
200 • 1 ms • 2.954 KB 1
PASS logout ends on login form and removes the cookie
GET Public - Display Landing Page
http://127.0.0.1:8000/
200 • 2 ms • 3.071 KB 1
PASS landing page loads
GET API - List Users
http://127.0.0.1:8000/api/users
200 • 3 ms • 435 B 1
PASS users API returns a JSON array